Tech Dessert #9: Using a Multimeter, and Soldering

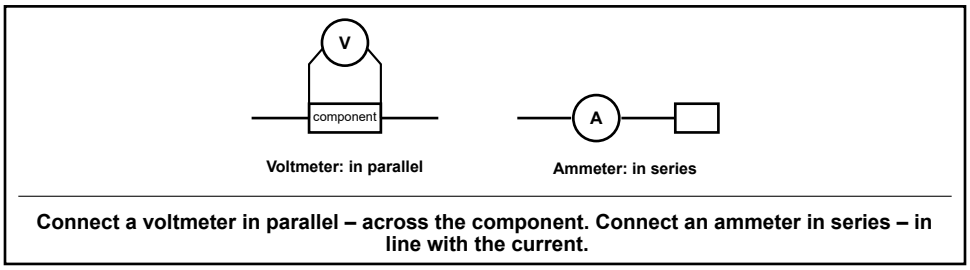
Companion to Bite #9 -- the test instruments every ham uses, how to connect them, and what makes a good solder joint.

Group T7D · 11 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert is the companion to Bite 9 The Bite measured the antenna system; this sheet covers the bench tools – the multimeter and the soldering iron. Read the table, study the diagram, then drill the questions.

Test Instruments and Soldering

Topic	What to know	Q
Instrument that measures voltage	A **voltmeter**.	T7D01
How a voltmeter connects	**In parallel** – across the component.	T7D02
How a meter connects to measure current	**In series** – in line with the current.	T7D03
Instrument that measures current	An **ammeter**.	T7D04
How an ohmmeter measures resistance	It applies a **small current** and measures the resulting voltage.	T7D05
What can damage a multimeter	Trying to measure **voltage** while it is set to **resistance**.	T7D06
What a multimeter measures	**Voltage and resistance** (and current).	T7D07
Solder you must NOT use for electronics	**Acid-core** solder.	T7D08
The look of a cold (bad) solder joint	A **rough or lumpy** surface.	T7D09
Ohmmeter across a large discharged capacitor	The resistance reading **increases with time** (the capacitor charges).	T7D10
Measuring resistance of a circuit	Make sure the circuit is **not powered**.	T7D11



Connect a voltmeter in parallel – across the component. Connect an ammeter in series – in line with the current.

Remember the one rule that matters most: a **voltmeter goes in parallel** (across a component) and an **ammeter goes in series** (in the current path). Set the multimeter to the right function before you connect it – measuring voltage on the resistance setting can damage it – and never measure resistance on a **powered** circuit. For soldering: use rosin-core, never **acid-core**, and a good joint is smooth and shiny, not rough and lumpy.

Dessert 9 Question Bank (T7D)

T7D01. Which instrument would you use to measure electric potential?
A. An ammeter
B. A voltmeter
C. A potentiometer
D. An ohmmeter

T7D02. How is a voltmeter connected to a component to measure applied voltage?
A. In series
B. In parallel
C. In quadrature
D. In phase

T7D03. When configured to measure current, how is a multimeter connected to a component?
A. In series
B. In parallel
C. In quadrature
D. In phase

T7D04. Which instrument is used to measure electric current?
A. An ohmmeter
B. An electrometer
C. A voltmeter
D. An ammeter

T7D05. How does an ohmmeter measure the resistance of a circuit or component?
A. By applying a small current and measuring the resulting voltage
B. By placing a variable resistor in parallel with the circuit
C. By placing a variable resistor in series with the circuit
D. By applying a variable voltage and measuring the resulting current change

T7D06. Which of the following can damage a multimeter?
A. Attempting to measure resistance using the voltage setting
B. Failing to connect one of the probes to ground
C. Attempting to measure voltage when using the resistance setting
D. Not allowing it to warm up properly

T7D07. Which of the following measurements are made using a multimeter?
A. Signal strength and noise
B. Impedance and reactance
C. Voltage and resistance
D. All these choices are correct

T7D08. Which of the following types of solder should not be used for radio and electronic applications?
A. Acid-core solder
B. Lead-tin solder
C. Rosin-core solder
D. Tin-copper solder

T7D09. What is the characteristic appearance of a cold tin-lead solder joint?
A. Dark black spots
B. A bright or shiny surface
C. A rough or lumpy surface
D. A greenish tinge

T7D10. What reading indicates that an ohmmeter is connected across a large, discharged capacitor?
A. Increasing resistance with time
B. Decreasing resistance with time
C. Steady full-scale reading
D. Alternating between open and short circuit

T7D11. Which of the following precautions should be taken when measuring in-circuit resistance with an ohmmeter?
A. Ensure that the applied voltages are correct
B. Ensure that the circuit is not powered
C. Ensure that the circuit is grounded
D. Ensure that the circuit is operating at the correct frequency

Answer Key

T7D01: B	T7D02: B	T7D03: A	T7D04: D
T7D05: A	T7D06: C	T7D07: C	T7D08: A
T7D09: C	T7D10: A	T7D11: B	